

Adrián Vidal Delgado

PhD Candidate in Mathematics

Institute of Mathematics · Universidad de Talca, Chile

RESEARCH AREAS

Orthogonal polynomials

Symmetric functions

Integrable systems

Stochastic processes

EDUCATION

PhD in Mathematics

Universidad de Talca, 2024–present

Candidate since March 2026. Expected

completion: March 2028. Advisor: Jan

Felipe van Diejen.

M.Sc. in Mathematics

Universidad de Talca, 2024

Advisor: Jan Felipe van Diejen.

B.Sc. in Mathematics

Universidad de La Habana, 2020

ACADEMIC SERVICE

Organizing Committee, *Summer School for Undergraduates in Mathematics* (2026), Universidad de Talca.

RESEARCH OUTPUTS

The Spectrum of the Finite Open XX Quantum Spin Chain with Transverse Magnetic Boundary Fields via Orthogonal Polynomials

J. F. van Diejen, A. Soledispa Tibán, **A. Vidal**

Annales Henri Poincaré 27 (2026), 1451–1495. DOI

Approximation of Askey–Wilson roots

J. F. van Diejen, A. Soledispa Tibán, **A. Vidal**

Accepted for publication in *Constructive Approximation*. arXiv:2501.17710

SELECTED PRESENTATIONS

AUGUST 2026 · COSTA RICA

Hall–Littlewood Polynomials and Interacting Random Walks on Partitions

CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry.

JUNE 2026 · SPAIN

Biased random walks on partitions and hyperoctahedral Hall–Littlewood polynomials

OPSF-S11 Summer School, Universidad de Alcalá.

APRIL 2026 · CHILE

Aproximación de las raíces de los polinomios de Askey–Wilson mediante una iteración de punto fijo

XXXVIII Jornada de Matemática de la Zona Sur, Universidad del Bío-Bío.

OCTOBER 2025 · ONLINE

Análisis espectral de una clase de cadenas de espín cuántico XX

ICMAM Central America and the Caribbean 2025.

TEACHING

Lecturer · Universidad de Talca (2025)

Mathematics for Engineering in Nutrition and Dietetics.

Teaching Assistant · Universidad de Talca (2022–2024)

Calculus, Linear Algebra, Probability and Statistics, and ODEs.

Earlier teaching · Havana (2018–2022)

Universidad de La Habana and CUJAE.